GPU-TRAINED EDUCATIONAL MODEL

Synthetic cataract grade CNN

Compact convolutional image classifier - four generated opacity grades

SYNTHETIC ONLY Every image, clinical value and target was generated procedurally. Results are not human annotations and do not measure performance on real patients.

Measured synthetic result

Metric	Held-out synthetic test
Samples	800
Accuracy	100.00%
Balanced accuracy	100.00%
Macro F1	100.00%
Best validation epoch	18
Training device	NVIDIA GeForce RTX 2050 (cuda)
Training time	103.2 seconds

Why the score is perfect

The procedural generator intentionally encodes separable opacity patterns and correlated fictional variables. A perfect held-out score shows reproducibility and successful pipeline integration; it is expected for this educational task and must not be generalized to real slit-lamp imaging.

CLINICAL CLAIM None. The model must remain inside the Simulation Lab and must never grade uploaded patient imagery.

TRAINING DIAGNOSTICS

Learning curves and error structure

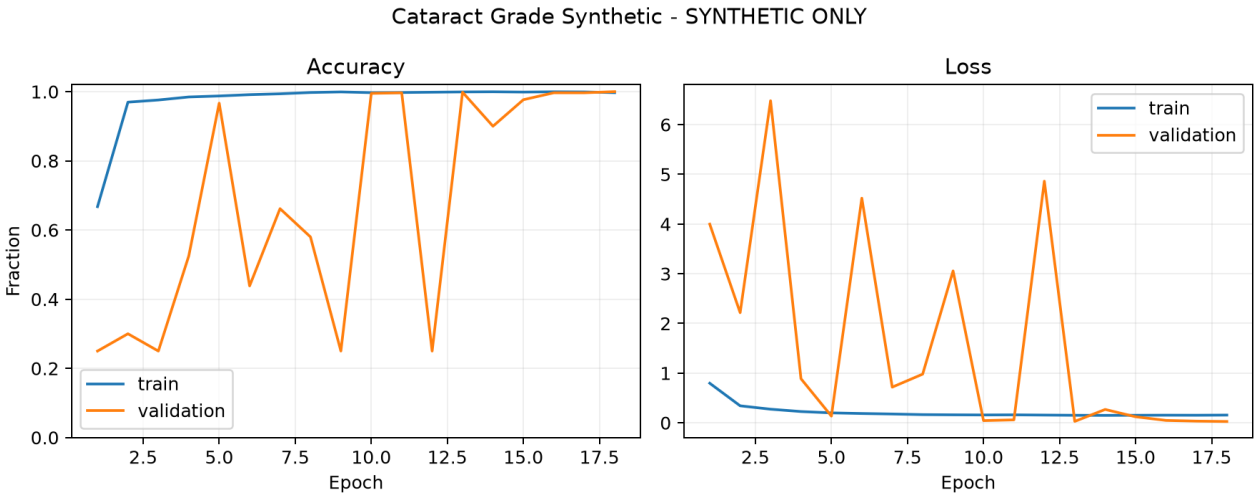


Figure 1. Training and validation traces. Instability in early validation reinforces that the selected checkpoint, not the final training state, is the evidence artifact.

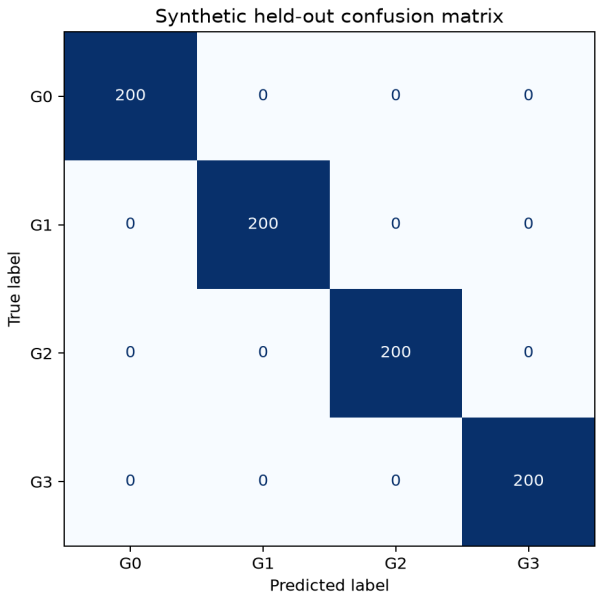


Figure 2. Held-out synthetic confusion matrix (200 generated samples per class).

Reproducibility boundary

Fixed random seed: 20261025. Four balanced grades. Training: 2,400 generated samples; validation: 600; test: 800. The test generator uses disjoint seeds. The checkpoint records the class order, epoch and synthetic-only flag.